

Instructors: Dr. Jim D. Pham and Paul Nguyen

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Office Hrs: Wed, 3 PM – 4PM (Canvas)

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Section: CNET-170 - Hybrid

COURSE DESCRIPTION:

This course provides an in-depth study of Network Security fundamentals and provides a comprehensive overview of network security. Students will gain the knowledge and skills required to identify risk and participate in risk mitigation activities; provide infrastructure, application, operational, and information security; apply security controls to maintain confidentiality, integrity, and availability; identify appropriate technologies and products; and operate with an awareness of applicable policies, laws, and regulations. This course provides the foundation for students preparing to take the CompTIA Security+ certification exam.

TEXTBOOK: CompTIA Security+ Guide to Network Security Fundamentals, 5th Edition By:
Mark Ciampa

ISBN-10: 1305093917; ISBN-13: 9781305093911

COURSE ASSIGNMENTS:

1. Reading Assignments: Textbook readings and online supporting webpages to inform the student on the fundamental concepts of computer security, security threats and vulnerabilities, and on how to examine network security.
2. Projects, Activities, and other Assignments: Hands-on lab assignments using the Ohlone NetLAB to develop skills on how to manage application, data and host security and identify access control and account management security measures. Troubleshooting non-expected outcomes.
3. Writing Assignments: Assignment worksheets and lab reports to support the lab assignments and document the results of those lab assignments.

STUDENT LEARNING OUTCOMES:

After completing the course, students will be able to:

1. Identify the fundamental concepts of computer security, security threats and vulnerabilities and examine network security.
2. Manage application, data and host security and identify access control and account management security measures.
3. Identify compliance and operational security measures.
4. Manage certificates, risk, and security incidents.
5. Develop business continuity and disaster recovery plans

TEACHING METHODS:

1. Lectures: The presentations slides and assignments will be accessed through Ohlone's Canvas Course Management System. You access the online resource at this URL:
<https://ohlone.instructure.com/>
2. See Instructional Resources below.
3. Laboratory: NetLAB and Research Project

DISCUSSION BOARD:

Your participation on the Discussion Board (on Canvas) will be an important aspect of the class. You will be required to post, and respond to at least two posts from a classmate, on a weekly basis.

INSTRUCTIONAL RESOURCES:

1. **Live Canvas Web Conferencing**
2. **Canvas:** Class documents, syllabus, discussion, and the lecture slides will be posted on Ohlone's Canvas. Canvas is an online Course Management System. You access Canvas at this URL:
<https://ohlone.instructure.com/>.

CANCAS WEB CONFERENCING:

I will be conducting weekly live lectures and labs through Canvas Conferences. You may remotely attend these live lectures or review the archives at a later date.

Schedule:

The following is a tentative in class meeting schedule. The instructor reserves the right to make any schedule changes deemed necessary. *Please see the detail schedule where it lists as Conference.*

STUDENT/GROUP-LED DISCUSSION:

Each student/group will be assigned a chapter and prepare an outline/paper (1-2 pages) for that chapter and PowerPoint 30 minutes presentation.

The discussion will provide the opportunity for the class to reflect on the topics covered in class, reading.

EMAIL: If you send me email please put **course number – name – topic** into the email **Subject** field. For example: CNET 170 – Name – Issue

LAB ASSIGNMENTS: There will be weekly lab assignments one or two per week, depending on the scope and difficulty of the assignment.

POLICY ON STUDENT WORK: The NetLAB assignments are to be done individually, unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

QUIZZES: There will be quizzes on the reading assignments and on the prior week's material. There are no makeups for quizzes.

GRADING:

Labs:	30%
Participation:	10%
Project:	20%
Quizzes:	10%
Final Exam:	30%

GRADING SCALE:

A = 90-100
B = 80-89
C = 65-79
D = 55-64
F = <55

COURSE SCHEDULE: The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Session	Date	Activity
1	8/27/18	Chapter 1: Introduction to Security (NC2309) Lab: Research Project Security
2	9/03/18	Holiday: Labor Day (Ohlone College closed)
3	9/10/18	Chapter 2: Malware and Social Engineering Attacks (Conference) Lab: Research Project Social Engineering
4	9/17/18	Chapter 3: Application and Networking-Based Attacks (NC2309) NetLAB: Lab 1: Network Devices and Technologies - Capturing Network Traffic Or Lab 02: Secure Network Administration Principles Log Analysis
5	9/24/18	Chapter 4: Host, Application, and Data Security (Conference) NetLAB: Lab 03: Protocols and Default Network Ports - Transferring Data Using TCP/IP Or Lab 04: Protocols and Default Network Ports - Connecting to a Remote System
6	10/01/18	Chapter 5: Basic Cryptography (Conference) NetLAB: Lab 19: General Cryptography Concepts Or Lab 20: Cryptography
7	10/08/18	Chapter 6: Advanced Cryptography (NC2309)
8	10/15/18	Chapter 7: Network Security (Conference) NetLAB: Lab 06: Incident Response Procedures Or Lab 07: Configuring the pfSense Firewall Or Lab 08: Configuring Backups
9	10/22/18	Chapter 8: Administering a Secure Network (Conference) NetLAB: Lab 09: Analyze and Differentiate Types of Malware Or Lab 10: Analyze and Differentiate Types of Attacks Using Windows Commands Or Lab 11: Analyze and Differentiate Types of Application Attacks
10	10/29/18	Chapter 9: Wireless Network Security (NC2309) NetLAB: Lab 05: Secure Implementation of Wireless Networking

	11/05/18	Chapter 10: Mobile Device Security (Conference) NetLab: Continuing Lab 05: Secure Implementation of Wireless Networking
12	11/12/18	Chapter 11: Access Control Fundamentals (Conference) NetLAB: Lab 17: Authentication, Authorization and Access Control Or Lab 18: Access Control/Audit
13	11/19/18	Chapter 12: Authentication and Account Management (Conference) NetLAB: Continuing Lab 17/18
14	11/26/18	Chapter 13: Business Continuity (Conference) NetLAB: Lab 12: Mitigation and Deterrent Techniques - Anti-Forensic Or Lab 13: Mitigation and Deterrent Techniques - Password Cracking
15	12/03/18	Chapter 14: Risk Mitigation (Conference) NetLAB: Continuing Lab 12/13
16	12/10/18	Chapter 15: Vulnerability Assessment and Third-Party Integration (NC2309) NetLAB: Lab 15: Importance of Data Security - Data Theft Or Lab 16: Importance of Data Security - Securing Data Using Encryption Software Final Review
17	12/11/18	Final Exam (ONLINE)